



# 1947 - JWST Survey of the Prototypical Core-collapse Supernova Remnant Cassiopeia A

Cycle: 1, Proposal Category: GO

## INVESTIGATORS

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| <i>Name</i>                                  | <i>Institution</i>                                | <i>E-Mail</i>                      |
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**OBSERVATIONS**

| <i>Folder</i>      | <i>Observation</i> | <i>Label</i>            | <i>Observing Template</i>           | <i>Science Target</i>         |
|--------------------|--------------------|-------------------------|-------------------------------------|-------------------------------|
| Observation Folder |                    |                         |                                     |                               |
|                    | 1                  | MIRI Imaging Mosaic     | MIRI Imaging                        | (10) CASSIOPEIA-A-CENTER-MIRI |
|                    | 2                  | MIRI Imaging Background | MIRI Imaging                        | (3) BACKGROUND                |
|                    | 3                  | MIRI IFU Center         | MIRI Medium Resolution Spectroscopy | (11) CASSIOPEIA-A-CENTER-IFU  |
|                    | 4                  | MIRI IFU Knot 1         | MIRI Medium Resolution Spectroscopy | (2) CASSIOPEIA-A-KNOT1        |
|                    | 5                  | MIRI IFU Knot 2         | MIRI Medium Resolution Spectroscopy | (8) CASSIOPEIA-A-KNOT2        |
|                    | 6                  | MIRI IFU Knot 3         | MIRI Medium Resolution Spectroscopy | (9) CASSIOPEIA-A-KNOT3        |
|                    | 7                  | MIRI IFU Background     | MIRI Medium Resolution Spectroscopy | (15) BACKGROUND-RETURN        |
|                    | 8                  | NIRSpec Knot 1          | NIRSpec IFU Spectroscopy            | (2) CASSIOPEIA-A-KNOT1        |
|                    | 9                  | NIRSpec Knot 2          | NIRSpec IFU Spectroscopy            | (8) CASSIOPEIA-A-KNOT2        |

| <b>Folder</b>       | <b>Observation</b> | <b>Label</b>                       | <b>Observing Template</b>       | <b>Science Target</b>        |
|---------------------|--------------------|------------------------------------|---------------------------------|------------------------------|
|                     | 10                 | NIRSpec Knot 3                     | NIRSpec IFU Spectroscopy        | (9) CASSIOPEIA-A-KNOT3       |
|                     | 11                 | NIRSpec Knot Bkg                   | NIRSpec IFU Spectroscopy        | (15) BACKGROUND-RETURN       |
|                     | 12                 | NIRSpec Center                     | NIRSpec Fixed Slit Spectroscopy | (11) CASSIOPEIA-A-CENTER-IFU |
|                     | 13                 | NIRCam imaging                     | NIRCam Imaging                  | (1) CASSIOPEIA-A-CENTER      |
| Additional coverage |                    |                                    |                                 |                              |
|                     | 15                 | MIRI Imaging additional 1          | MIRI Imaging                    | (12) CASSIOPEIA-A-1-RETURN   |
|                     | 16                 | MIRI Imaging additional 2          | MIRI Imaging                    | (13) CASSIOPEIA-A-2-RETURN   |
|                     | 17                 | MIRI Imaging additional 3          | MIRI Imaging                    | (14) CASSIOPEIA-A-3-RETURN   |
|                     | 18                 | MIRI Imaging additional background | MIRI Imaging                    | (15) BACKGROUND-RETURN       |

## ABSTRACT

Core-collapse supernovae (SNe) are among the most influential phenomena in the universe, and yet many key questions about their nature and explosive products remain unanswered. This uncertainty has broad implications for the formation and evolution of stellar populations, the metal enrichment of galaxies, and the origin of life. JWST opens new pathways to investigate SNe, and this proposal outlines an interdisciplinary science-enabling survey of the young SN remnant that provides the clearest access to the properties of a core-collapse SN: Cassiopeia A (Cas A). Three critical questions motivate a suite of observations made up of imaging mosaics covering the entire main shell and IFU spectroscopy of select representative locations: What is the total mass, relative chemical yield, and kinematic distribution of various components of the SN ejecta? How much ejecta is transformed into dust and how much of that dust survives passage through the reverse shock? What processes govern the formation and final fate of the remnant compact object? The requested observations exploit JWST's unique ability to 1) provide maps of shocked and unshocked ejecta that can be directly compared to current SN models; 2) constrain the grain size distribution, clump size, and density contrast of shocked SN dust; and 3) peer deeply enough through high extinction to test hypotheses about the nature of Cas A's central compact object, which is regarded as a key object to understanding neutron star evolution models. The proposed program will serve as an invaluable resource for subsequent JWST cycles and will contribute to the legacy of prior NASA Great Observatories that have all observed Cas A.

## OBSERVING DESCRIPTION

A suite of imaging mosaics with NIRCam + MIRI covering the entire main shell of the supernova remnant Cassiopeia A (Cas A) will be obtained. NIRSpec + MIRI MRS spectroscopy of four representative locations will also be obtained. Three locations will sample main shell ejecta and dust

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with NIRSpec IFU and MIRI MRS IFU. One "deep drilling" location near the center will sample unshocked ejecta with NIRSpec + long slit, and an extra-long MIRI MRS IFU observation. NIRSpec observations will all use the G395M disperser, and all MIRI MRS observations will span the full 5-28 micron range over multiple exposures. All IFU observations will be dithered.

NIRCam filters F162M, F356W, and F444W have been selected to map ejecta, dust, and CSM with emission lines of [Fe II], [Si I], and CO. Images from these filters will also be used to constrain emission from the CCO. MIRI filters F560W, F770W, F1000W, F1130W, F1280W, F1800W, F2100W, and F2550W have been selected to map ejecta and dust continuum for a variety of emission features. Mosaic configurations have been carefully selected to optimally cover the remnant's main shell, base of NE jet, and outlying ejecta + CSM.

In Cas A, knots move with proper motions upwards of 0.9 arcseconds per year and flicker on timescales of months (see Fesen+ 2011), and thus we require turnaround between imaging and spectroscopy to be within a single cycle. NIRSpec and MIRI IFU observations are to be obtained after NIRCam and MIRI imaging has been obtained. Precise coordinates of NIRSpec and MIRI MRS observations will be made from NIRCam and MIRI images. This requirement is reflected in an observational constraint that the imaging observations with MIRI and NIRCam should be carried out before the IFU observations of the fast moving knots in order for us to be able to modify the target coordinates if needed. We specified a timing window for this constraint of 60-180 days. The 60 days is an estimate of the time needed for the pipeline to process the imaging observations, for reduce the data, plus the time needed to schedule the observation after the updated APT file is submitted.

We do not require a dedicated spectral background observation for the CENTER-IFU target. We are only interested in line emission at this location (and not the continuum) and any lines associated with the remnant will be shifted in velocity from the background lines and easily identifiable. We will therefore only need to fit the underlying continuum to measure the line properties without the need for any off-remnant background.

# Proposal 1947 - Targets - JWST Survey of the Prototypical Core-collapse Supernova Remnant Cassiopeia A

| Fixed Targets | #    | Name                     | Target Coordinates                                                                                                                                                                                                                                                                                                                                                                   | Targ. Coord. Corrections | Miscellaneous |
|---------------|------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|---------------|
|               | (1)  | CASSIOPEIA-A-CENTER      | RA: 23 23 24.0000 (350.8500000d)<br>Dec: +58 48 54.00 (58.81500d)<br>Equinox: J2000<br><br><i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. This object was generated by the targetselector and retrieved from the SIMBAD database.</i><br><i>Category=ISM</i><br><i>Description=[Supernova remnants]</i><br><i>Extended=YES</i> |                          |               |
|               | (2)  | CASSIOPEIA-A-KNOT1       | RA: 23 23 29.5212 (350.8730050d)<br>Dec: +58 50 31.34 (58.84204d)<br>Equinox: J2000<br><br><i>Comments:</i><br><i>Category=ISM</i><br><i>Description=[Supernova remnants]</i><br><i>Extended=YES</i>                                                                                                                                                                                 |                          |               |
|               | (3)  | BACKGROUND               | RA: 23 22 44.6340 (350.6859750d)<br>Dec: +58 51 42.19 (58.86172d)<br>Equinox: J2000<br><br><i>Comments:</i><br><i>Category=Calibration</i><br><i>Description=[Telescope/sky background]</i><br><i>Extended=YES</i>                                                                                                                                                                   |                          |               |
|               | (8)  | CASSIOPEIA-A-KNOT2       | RA: 23 23 25.9879 (350.8582829d)<br>Dec: +58 48 43.46 (58.81207d)<br>Equinox: J2000<br><br><i>Comments:</i><br><i>Category=ISM</i><br><i>Description=[Supernova remnants]</i><br><i>Extended=YES</i>                                                                                                                                                                                 |                          |               |
|               | (9)  | CASSIOPEIA-A-KNOT3       | RA: 23 23 30.3311 (350.8763796d)<br>Dec: +58 47 25.02 (58.79028d)<br>Equinox: J2000<br><br><i>Comments:</i><br><i>Category=ISM</i><br><i>Description=[Supernova remnants]</i><br><i>Extended=YES</i>                                                                                                                                                                                 |                          |               |
|               | (10) | CASSIOPEIA-A-CENTER-MIRI | RA: 23 23 27.3589 (350.8639954d)<br>Dec: +58 49 8.42 (58.81901d)<br>Equinox: J2000<br><br><i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. This object was generated by the targetselector and retrieved from the SIMBAD database.</i><br><i>Category=ISM</i><br><i>Description=[Supernova remnants]</i><br><i>Extended=YES</i>  |                          |               |
|               | (11) | CASSIOPEIA-A-CENTER-IFU  | RA: 23 23 27.8999 (350.8662496d)<br>Dec: +58 48 37.88 (58.81052d)<br>Equinox: J2000<br><br><i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. This object was generated by the targetselector and retrieved from the SIMBAD database.</i><br><i>Category=ISM</i><br><i>Description=[Supernova remnants]</i><br><i>Extended=YES</i> |                          |               |

## Proposal 1947 - Targets - JWST Survey of the Prototypical Core-collapse Supernova Remnant Cassiopeia A

|      |                       |                                                                                                                                                                                                                                                                                                                                                                                      |
|------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (12) | CASSIOPEIA-A-1-RETURN | RA: 23 23 20.7612 (350.8365050d)<br>Dec: +58 47 37.42 (58.79373d)<br>Equinox: J2000<br><br><i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. This object was generated by the targetselector and retrieved from the SIMBAD database.</i><br><i>Category=ISM</i><br><i>Description=[Supernova remnants]</i><br><i>Extended=YES</i> |
| (13) | CASSIOPEIA-A-2-RETURN | RA: 23 23 32.8003 (350.8866679d)<br>Dec: +58 48 32.35 (58.80899d)<br>Equinox: J2000<br><br><i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. This object was generated by the targetselector and retrieved from the SIMBAD database.</i><br><i>Category=ISM</i><br><i>Description=[Supernova remnants]</i><br><i>Extended=YES</i> |
| (14) | CASSIOPEIA-A-3-RETURN | RA: 23 23 33.7392 (350.8905800d)<br>Dec: +58 50 21.01 (58.83917d)<br>Equinox: J2000<br><br><i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. This object was generated by the targetselector and retrieved from the SIMBAD database.</i><br><i>Category=ISM</i><br><i>Description=[Supernova remnants]</i><br><i>Extended=YES</i> |
| (15) | BACKGROUND-RETURN     | RA: 23 24 19.8259 (351.0826079d)<br>Dec: +58 43 7.26 (58.71868d)<br>Equinox: J2000<br><br><i>Comments:</i><br><i>Category=Calibration</i><br><i>Description=[Telescope/sky background]</i><br><i>Extended=YES</i>                                                                                                                                                                    |

# Proposal 1947 - Observation 1 - JWST Survey of the Prototypical Core-collapse Supernova Remnant Cassiopeia A

|                   |                                                                                                                                                                                          |                          |                                                                                    |                  |                  |                          |                |               |                    |                     |                              |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|------------------------------------------------------------------------------------|------------------|------------------|--------------------------|----------------|---------------|--------------------|---------------------|------------------------------|
| Observation       | Proposal 1947, Observation 1: MIRI Imaging Mosaic                                                                                                                                        |                          |                                                                                    |                  |                  |                          |                |               |                    |                     | Thu Nov 10 23:00:46 GMT 2022 |
|                   | Diagnostic Status: Warning                                                                                                                                                               |                          |                                                                                    |                  |                  |                          |                |               |                    |                     |                              |
|                   | Observing Template: MIRI Imaging                                                                                                                                                         |                          |                                                                                    |                  |                  |                          |                |               |                    |                     |                              |
|                   | Background Observations:[MIRI Imaging Background (Obs 2)]                                                                                                                                |                          |                                                                                    |                  |                  |                          |                |               |                    |                     |                              |
| Diagnostics       | (MIRI Imaging Mosaic (Obs 1)) Warning (Form): Use of background targets not expected with this template                                                                                  |                          |                                                                                    |                  |                  |                          |                |               |                    |                     |                              |
|                   | (Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                              |                          |                                                                                    |                  |                  |                          |                |               |                    |                     |                              |
|                   | (Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                              |                          |                                                                                    |                  |                  |                          |                |               |                    |                     |                              |
|                   | (Visit 1:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                              |                          |                                                                                    |                  |                  |                          |                |               |                    |                     |                              |
|                   | (Visit 1:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                              |                          |                                                                                    |                  |                  |                          |                |               |                    |                     |                              |
|                   | (Visit 1:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                              |                          |                                                                                    |                  |                  |                          |                |               |                    |                     |                              |
|                   | (Visit 1:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                              |                          |                                                                                    |                  |                  |                          |                |               |                    |                     |                              |
|                   | (Visit 1:7) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                              |                          |                                                                                    |                  |                  |                          |                |               |                    |                     |                              |
|                   | (Visit 1:8) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                              |                          |                                                                                    |                  |                  |                          |                |               |                    |                     |                              |
|                   | (Visit 1:9) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                              |                          |                                                                                    |                  |                  |                          |                |               |                    |                     |                              |
|                   | (Visit 1:10) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                             |                          |                                                                                    |                  |                  |                          |                |               |                    |                     |                              |
| Fixed Targets     | #                                                                                                                                                                                        | Name                     | Target Coordinates                                                                 |                  |                  | Targ. Coord. Corrections |                |               | Miscellaneous      |                     |                              |
|                   | (10)                                                                                                                                                                                     | CASSIOPEIA-A-CENTER-MIRI | RA: 23 23 27.3589 (350.8639954d)<br>Dec: +58 49 8.42 (58.81901d)<br>Equinox: J2000 |                  |                  |                          |                |               |                    |                     |                              |
|                   | Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.This object was generated by the targetselector and retrieved from the SIMBAD database. |                          |                                                                                    |                  |                  |                          |                |               |                    |                     |                              |
|                   | Category=ISM<br>Description=[Supernova remnants]<br>Extended=YES                                                                                                                         |                          |                                                                                    |                  |                  |                          |                |               |                    |                     |                              |
| Template          | Subarray                                                                                                                                                                                 |                          |                                                                                    |                  |                  |                          |                |               |                    |                     |                              |
|                   | FULL                                                                                                                                                                                     |                          |                                                                                    |                  |                  |                          |                |               |                    |                     |                              |
| Mosaic            | Rows                                                                                                                                                                                     | Columns                  | Row Overlap %                                                                      |                  | Column Overlap % |                          | Row shift      | Column shift  |                    | Tile Order          |                              |
|                   | 3                                                                                                                                                                                        | 5                        | 10.0                                                                               |                  | 10.0             |                          | 0.0            | 0.0           |                    | DEFAULT             |                              |
| Dithers           | #                                                                                                                                                                                        | Dither Type              | Starting Point                                                                     | Number of Points | Points           | Starting Set             | Number of Sets | Optimized For | Direction          | Pattern Size        |                              |
|                   | 1                                                                                                                                                                                        | 4-Point-Sets             | 1                                                                                  | 4                |                  | 6                        | 1              | POINT SOURCE  | POSITIVE           | DEFAULT             |                              |
| Spectral Elements | #                                                                                                                                                                                        | Filter                   | Readout Pattern                                                                    | Groups/Int       | Integrations/Exp | Exposures/Dith           | Dither         | Total Dithers | Total Integrations | Total Exposure Time | ETC Wkbk.Calc ID             |
|                   | 1                                                                                                                                                                                        | F560W                    | FASTR1                                                                             | 8                | 1                | 1                        | Dither 1       | 4             | 4                  | 88.801              |                              |
|                   | 2                                                                                                                                                                                        | F770W                    | FASTR1                                                                             | 8                | 1                | 1                        | Dither 1       | 4             | 4                  | 88.801              |                              |
|                   | 3                                                                                                                                                                                        | F1000W                   | FASTR1                                                                             | 8                | 1                | 1                        | Dither 1       | 4             | 4                  | 88.801              |                              |
|                   | 4                                                                                                                                                                                        | F1130W                   | FASTR1                                                                             | 8                | 1                | 1                        | Dither 1       | 4             | 4                  | 88.801              |                              |
|                   | 5                                                                                                                                                                                        | F1280W                   | FASTR1                                                                             | 8                | 1                | 1                        | Dither 1       | 4             | 4                  | 88.801              |                              |
|                   | 6                                                                                                                                                                                        | F1800W                   | FASTR1                                                                             | 8                | 1                | 1                        | Dither 1       | 4             | 4                  | 88.801              |                              |
|                   | 7                                                                                                                                                                                        | F2100W                   | FASTR1                                                                             | 8                | 1                | 1                        | Dither 1       | 4             | 4                  | 88.801              |                              |
|                   | 8                                                                                                                                                                                        | F2550W                   | FASTR1                                                                             | 8                | 1                | 1                        | Dither 1       | 4             | 4                  | 88.801              |                              |

Proposal 1947 - Observation 1 - JWST Survey of the Prototypical Core-collapse Supernova Remnant Cassiopeia A

|                      |                                                                                                      |
|----------------------|------------------------------------------------------------------------------------------------------|
| Special Requirements | Group Visits within 53.0 Days<br>Visits Same PA<br><br>Sequence Observations 1, 2, Non-interruptible |
|----------------------|------------------------------------------------------------------------------------------------------|



# Proposal 1947 - Observation 2 - JWST Survey of the Prototypical Core-collapse Supernova Remnant Cassiopeia A

|                      |                                                                                             |              |                                  |                  |                  |                          |                |               |                    |                     |                              |
|----------------------|---------------------------------------------------------------------------------------------|--------------|----------------------------------|------------------|------------------|--------------------------|----------------|---------------|--------------------|---------------------|------------------------------|
| Observation          | Proposal 1947, Observation 2: MIRI Imaging Background                                       |              |                                  |                  |                  |                          |                |               |                    |                     | Thu Nov 10 23:00:46 GMT 2022 |
|                      | Diagnostic Status: Warning                                                                  |              |                                  |                  |                  |                          |                |               |                    |                     |                              |
|                      | Observing Template: MIRI Imaging                                                            |              |                                  |                  |                  |                          |                |               |                    |                     |                              |
|                      | Background Observation For: [MIRI Imaging Mosaic (Obs 1)]                                   |              |                                  |                  |                  |                          |                |               |                    |                     |                              |
| Diagnostics          | (Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. |              |                                  |                  |                  |                          |                |               |                    |                     |                              |
| Fixed Targets        | #                                                                                           | Name         | Target Coordinates               |                  |                  | Targ. Coord. Corrections |                |               | Miscellaneous      |                     |                              |
|                      | (3)                                                                                         | BACKGROUND   | RA: 23 22 44.6340 (350.6859750d) |                  |                  |                          |                |               |                    |                     |                              |
|                      |                                                                                             |              | Dec: +58 51 42.19 (58.86172d)    |                  |                  |                          |                |               |                    |                     |                              |
|                      |                                                                                             |              | Equinox: J2000                   |                  |                  |                          |                |               |                    |                     |                              |
| Template             | Comments:<br>Category=Calibration<br>Description=[Telescope/sky background]<br>Extended=YES |              |                                  |                  |                  |                          |                |               |                    |                     |                              |
|                      | Subarray                                                                                    |              |                                  |                  |                  |                          |                |               |                    |                     |                              |
|                      | FULL                                                                                        |              |                                  |                  |                  |                          |                |               |                    |                     |                              |
| Dithers              | #                                                                                           | Dither Type  | Starting Point                   | Number of Points | Points           | Starting Set             | Number of Sets | Optimized For | Direction          | Pattern Size        |                              |
|                      | 1                                                                                           | 4-Point-Sets | 1                                | 4                |                  | 6                        | 1              | POINT SOURCE  | POSITIVE           | DEFAULT             |                              |
| Spectral Elements    | #                                                                                           | Filter       | Readout Pattern                  | Groups/Int       | Integrations/Exp | Exposures/Dith           | Dither         | Total Dithers | Total Integrations | Total Exposure Time | ETC Wkbk.Calc ID             |
|                      | 1                                                                                           | F560W        | FASTR1                           | 8                | 1                | 1                        | Dither 1       | 4             | 4                  | 88.801              |                              |
|                      | 2                                                                                           | F770W        | FASTR1                           | 8                | 1                | 1                        | Dither 1       | 4             | 4                  | 88.801              |                              |
|                      | 3                                                                                           | F1000W       | FASTR1                           | 8                | 1                | 1                        | Dither 1       | 4             | 4                  | 88.801              |                              |
|                      | 4                                                                                           | F1130W       | FASTR1                           | 8                | 1                | 1                        | Dither 1       | 4             | 4                  | 88.801              |                              |
|                      | 5                                                                                           | F1280W       | FASTR1                           | 8                | 1                | 1                        | Dither 1       | 4             | 4                  | 88.801              |                              |
|                      | 6                                                                                           | F1800W       | FASTR1                           | 8                | 1                | 1                        | Dither 1       | 4             | 4                  | 88.801              |                              |
|                      | 7                                                                                           | F2100W       | FASTR1                           | 8                | 1                | 1                        | Dither 1       | 4             | 4                  | 88.801              |                              |
| 8                    | F2550W                                                                                      | FASTR1       | 8                                | 1                | 1                | Dither 1                 | 4              | 4             | 88.801             |                     |                              |
| Special Requirements | Sequence Observations 1, 2, Non-interruptible                                               |              |                                  |                  |                  |                          |                |               |                    |                     |                              |

# Proposal 1947 - Observation 3 - JWST Survey of the Prototypical Core-collapse Supernova Remnant Cassiopeia A

|                   |                                                                                                                                                                                                                                                              |                         |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-------------------------------------------------------------------------------------|--------|-----------------|----------------------|--------------------------|----------------|----------|-----------------|--------------------|---------------------|------------------------------|
| Observation       | Proposal 1947, Observation 3: MIRI IFU Center                                                                                                                                                                                                                |                         |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     | Thu Nov 10 23:00:46 GMT 2022 |
|                   | Diagnostic Status: Warning                                                                                                                                                                                                                                   |                         |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |
|                   | Observing Template: MIRI Medium Resolution Spectroscopy                                                                                                                                                                                                      |                         |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |
| Diagnostics       | (MIRI IFU Center (Obs 3)) Warning (Form): Imager Filter overlap.                                                                                                                                                                                             |                         |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |
|                   | (Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                                                  |                         |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |
| Fixed Targets     | #                                                                                                                                                                                                                                                            | Name                    | Target Coordinates                                                                  |        |                 |                      | Targ. Coord. Corrections |                |          | Miscellaneous   |                    |                     |                              |
|                   | (11)                                                                                                                                                                                                                                                         | CASSIOPEIA-A-CENTER-IFU | RA: 23 23 27.8999 (350.8662496d)<br>Dec: +58 48 37.88 (58.81052d)<br>Equinox: J2000 |        |                 |                      |                          |                |          |                 |                    |                     |                              |
|                   | Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.This object was generated by the targetselector and retrieved from the SIMBAD database.<br>Category=ISM<br>Description=[Supernova remnants]<br>Extended=YES |                         |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |
| Acquisition       | #                                                                                                                                                                                                                                                            | Target                  |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |
|                   | 1                                                                                                                                                                                                                                                            | NONE                    |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |
| Template          | AcqFilter                                                                                                                                                                                                                                                    | Primary Channel         |                                                                                     |        |                 | Simultaneous Imaging |                          |                |          | Imager Subarray |                    |                     |                              |
|                   | F560W                                                                                                                                                                                                                                                        | ALL                     |                                                                                     |        |                 | YES                  |                          |                |          | FULL            |                    |                     |                              |
| Dithers           | #                                                                                                                                                                                                                                                            | Dither Type             |                                                                                     |        |                 | Optimized For        |                          |                |          | Direction       |                    |                     |                              |
|                   | 1                                                                                                                                                                                                                                                            | 4-Point                 |                                                                                     |        |                 | EXTENDED SOURCE      |                          |                |          | NEGATIVE        |                    |                     |                              |
| Spectral Elements | #                                                                                                                                                                                                                                                            | Wavelength Range        | Detector                                                                            | Filter | Readout Pattern | Groups/Int           | Integrations/Exp         | Exposures/Dith | Dither   | Total Dithers   | Total Integrations | Total Exposure Time | ETC Wkbk.Calc ID             |
|                   | 1                                                                                                                                                                                                                                                            |                         | IMAGER                                                                              | F1130W | FASTR1          | 19                   | 5                        | 1              | Dither 1 | 4               | 20                 | 1098.916            |                              |
|                   | 1                                                                                                                                                                                                                                                            | SHORT(A)                | MRSLONG                                                                             |        | FASTR1          | 100                  | 1                        | 1              | Dither 1 | 4               | 4                  | 1110.016            |                              |
|                   | 1                                                                                                                                                                                                                                                            | SHORT(A)                | MRSSHORT                                                                            |        | FASTR1          | 100                  | 1                        | 1              | Dither 1 | 4               | 4                  | 1110.016            |                              |
|                   | 2                                                                                                                                                                                                                                                            |                         | IMAGER                                                                              | F1130W | FASTR1          | 19                   | 5                        | 1              | Dither 1 | 4               | 20                 | 1098.916            |                              |
|                   | 2                                                                                                                                                                                                                                                            | MEDIUM(B)               | MRSLONG                                                                             |        | FASTR1          | 100                  | 1                        | 1              | Dither 1 | 4               | 4                  | 1110.016            |                              |
|                   | 2                                                                                                                                                                                                                                                            | MEDIUM(B)               | MRSSHORT                                                                            |        | FASTR1          | 100                  | 1                        | 1              | Dither 1 | 4               | 4                  | 1110.016            |                              |
|                   | 3                                                                                                                                                                                                                                                            |                         | IMAGER                                                                              | F1500W | FASTR1          | 19                   | 5                        | 1              | Dither 1 | 4               | 20                 | 1098.916            |                              |
|                   | 3                                                                                                                                                                                                                                                            | LONG(C)                 | MRSLONG                                                                             |        | FASTR1          | 100                  | 1                        | 1              | Dither 1 | 4               | 4                  | 1110.016            |                              |
|                   | 3                                                                                                                                                                                                                                                            | LONG(C)                 | MRSSHORT                                                                            |        | FASTR1          | 100                  | 1                        | 1              | Dither 1 | 4               | 4                  | 1110.016            |                              |
|                   |                                                                                                                                                                                                                                                              |                         |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |

# Proposal 1947 - Observation 4 - JWST Survey of the Prototypical Core-collapse Supernova Remnant Cassiopeia A

|                   |                                                                                             |                    |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
|-------------------|---------------------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------|--------|-----------------|----------------------|--------------------------|----------------|----------|-----------------|--------------------|---------------------|------------------------------|--|
| Observation       | Proposal 1947, Observation 4: MIRI IFU Knot 1                                               |                    |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     | Thu Nov 10 23:00:46 GMT 2022 |  |
|                   | Diagnostic Status: Warning                                                                  |                    |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
|                   | Observing Template: MIRI Medium Resolution Spectroscopy                                     |                    |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
|                   | Background Observations:[MIRI IFU Background (Obs 7)]                                       |                    |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
| Diagnostics       | (MIRI IFU Knot 1 (Obs 4)) Warning (Form): Imager Filter overlap.                            |                    |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
|                   | (Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                    |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
| Fixed Targets     | #                                                                                           | Name               | Target Coordinates                                                                  |        |                 |                      | Targ. Coord. Corrections |                |          |                 | Miscellaneous      |                     |                              |  |
|                   | (2)                                                                                         | CASSIOPEIA-A-KNOT1 | RA: 23 23 29.5212 (350.8730050d)<br>Dec: +58 50 31.34 (58.84204d)<br>Equinox: J2000 |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
|                   | Comments:<br>Category=ISM<br>Description=[Supernova remnants]<br>Extended=YES               |                    |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
|                   |                                                                                             |                    |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
| Acquisition       | #                                                                                           | Target             |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
|                   | 1                                                                                           | NONE               |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
| Template          | AcqFilter                                                                                   | Primary Channel    |                                                                                     |        |                 | Simultaneous Imaging |                          |                |          | Imager Subarray |                    |                     |                              |  |
|                   | F560W                                                                                       | ALL                |                                                                                     |        |                 | YES                  |                          |                |          | FULL            |                    |                     |                              |  |
| Dithers           | #                                                                                           | Dither Type        |                                                                                     |        |                 | Optimized For        |                          |                |          | Direction       |                    |                     |                              |  |
|                   | 1                                                                                           | 4-Point            |                                                                                     |        |                 | EXTENDED SOURCE      |                          |                |          | NEGATIVE        |                    |                     |                              |  |
| Spectral Elements | #                                                                                           | Wavelength Range   | Detector                                                                            | Filter | Readout Pattern | Groups/Int           | Integrations/Exp         | Exposures/Dith | Dither   | Total Dithers   | Total Integrations | Total Exposure Time | ETC Wkbk.Calc ID             |  |
|                   | 1                                                                                           |                    | IMAGER                                                                              | F1500W | FASTR1          | 8                    | 5                        | 1              | Dither 1 | 4               | 20                 | 488.407             |                              |  |
|                   | 1                                                                                           | LONG(C)            | MRSLONG                                                                             |        | FASTR1          | 50                   | 1                        | 1              | Dither 1 | 4               | 4                  | 555.008             |                              |  |
|                   | 1                                                                                           | LONG(C)            | MRSSHORT                                                                            |        | FASTR1          | 50                   | 1                        | 1              | Dither 1 | 4               | 4                  | 555.008             |                              |  |
|                   | 2                                                                                           |                    | IMAGER                                                                              | F770W  | FASTR1          | 8                    | 5                        | 1              | Dither 1 | 4               | 20                 | 488.407             |                              |  |
|                   | 2                                                                                           | MEDIUM(B)          | MRSLONG                                                                             |        | FASTR1          | 50                   | 1                        | 1              | Dither 1 | 4               | 4                  | 555.008             |                              |  |
|                   | 2                                                                                           | MEDIUM(B)          | MRSSHORT                                                                            |        | FASTR1          | 50                   | 1                        | 1              | Dither 1 | 4               | 4                  | 555.008             |                              |  |
|                   | 3                                                                                           |                    | IMAGER                                                                              | F560W  | FASTR1          | 8                    | 5                        | 1              | Dither 1 | 4               | 20                 | 488.407             |                              |  |
|                   | 3                                                                                           | SHORT(A)           | MRSLONG                                                                             |        | FASTR1          | 50                   | 1                        | 1              | Dither 1 | 4               | 4                  | 555.008             |                              |  |
|                   | 3                                                                                           | SHORT(A)           | MRSSHORT                                                                            |        | FASTR1          | 50                   | 1                        | 1              | Dither 1 | 4               | 4                  | 555.008             |                              |  |
|                   |                                                                                             |                    |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
|                   |                                                                                             |                    |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
|                   |                                                                                             |                    |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |

Proposal 1947 - Observation 4 - JWST Survey of the Prototypical Core-collapse Supernova Remnant Cassiopeia A

|                      |                                                                                                                    |
|----------------------|--------------------------------------------------------------------------------------------------------------------|
| Special Requirements | Between Dates 03-OCT-2022:00:00:00 and 01-FEB-2023:00:00:00<br>Sequence Observations 4, 5, 6, 7, Non-interruptible |
|----------------------|--------------------------------------------------------------------------------------------------------------------|

# Proposal 1947 - Observation 5 - JWST Survey of the Prototypical Core-collapse Supernova Remnant Cassiopeia A

|                   |                                                                                             |                    |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
|-------------------|---------------------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------|--------|-----------------|----------------------|--------------------------|----------------|----------|-----------------|--------------------|---------------------|------------------------------|--|
| Observation       | Proposal 1947, Observation 5: MIRI IFU Knot 2                                               |                    |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     | Thu Nov 10 23:00:46 GMT 2022 |  |
|                   | Diagnostic Status: Warning                                                                  |                    |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
|                   | Observing Template: MIRI Medium Resolution Spectroscopy                                     |                    |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
|                   | Background Observations:[MIRI IFU Background (Obs 7)]                                       |                    |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
| Diagnostics       | (MIRI IFU Knot 2 (Obs 5)) Warning (Form): Imager Filter overlap.                            |                    |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
|                   | (Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                    |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
| Fixed Targets     | #                                                                                           | Name               | Target Coordinates                                                                  |        |                 |                      | Targ. Coord. Corrections |                |          |                 | Miscellaneous      |                     |                              |  |
|                   | (8)                                                                                         | CASSIOPEIA-A-KNOT2 | RA: 23 23 25.9879 (350.8582829d)<br>Dec: +58 48 43.46 (58.81207d)<br>Equinox: J2000 |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
|                   | Comments:<br>Category=ISM<br>Description=[Supernova remnants]<br>Extended=YES               |                    |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
|                   |                                                                                             |                    |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
| Acquisition       | #                                                                                           | Target             |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
|                   | 1                                                                                           | NONE               |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
| Template          | AcqFilter                                                                                   | Primary Channel    |                                                                                     |        |                 | Simultaneous Imaging |                          |                |          | Imager Subarray |                    |                     |                              |  |
|                   | F560W                                                                                       | ALL                |                                                                                     |        |                 | YES                  |                          |                |          | FULL            |                    |                     |                              |  |
| Dithers           | #                                                                                           | Dither Type        |                                                                                     |        |                 | Optimized For        |                          |                |          | Direction       |                    |                     |                              |  |
|                   | 1                                                                                           | 4-Point            |                                                                                     |        |                 | EXTENDED SOURCE      |                          |                |          | NEGATIVE        |                    |                     |                              |  |
| Spectral Elements | #                                                                                           | Wavelength Range   | Detector                                                                            | Filter | Readout Pattern | Groups/Int           | Integrations/Exp         | Exposures/Dith | Dither   | Total Dithers   | Total Integrations | Total Exposure Time | ETC Wkbk.Calc ID             |  |
|                   | 1                                                                                           |                    | IMAGER                                                                              | F1500W | FASTR1          | 8                    | 5                        | 1              | Dither 1 | 4               | 20                 | 488.407             |                              |  |
|                   | 1                                                                                           | LONG(C)            | MRSLONG                                                                             |        | FASTR1          | 50                   | 1                        | 1              | Dither 1 | 4               | 4                  | 555.008             |                              |  |
|                   | 1                                                                                           | LONG(C)            | MRSSHORT                                                                            |        | FASTR1          | 50                   | 1                        | 1              | Dither 1 | 4               | 4                  | 555.008             |                              |  |
|                   | 2                                                                                           |                    | IMAGER                                                                              | F770W  | FASTR1          | 8                    | 5                        | 1              | Dither 1 | 4               | 20                 | 488.407             |                              |  |
|                   | 2                                                                                           | MEDIUM(B)          | MRSLONG                                                                             |        | FASTR1          | 50                   | 1                        | 1              | Dither 1 | 4               | 4                  | 555.008             |                              |  |
|                   | 2                                                                                           | MEDIUM(B)          | MRSSHORT                                                                            |        | FASTR1          | 50                   | 1                        | 1              | Dither 1 | 4               | 4                  | 555.008             |                              |  |
|                   | 3                                                                                           |                    | IMAGER                                                                              | F560W  | FASTR1          | 8                    | 5                        | 1              | Dither 1 | 4               | 20                 | 488.407             |                              |  |
|                   | 3                                                                                           | SHORT(A)           | MRSLONG                                                                             |        | FASTR1          | 50                   | 1                        | 1              | Dither 1 | 4               | 4                  | 555.008             |                              |  |
|                   | 3                                                                                           | SHORT(A)           | MRSSHORT                                                                            |        | FASTR1          | 50                   | 1                        | 1              | Dither 1 | 4               | 4                  | 555.008             |                              |  |
|                   |                                                                                             |                    |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
|                   |                                                                                             |                    |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
|                   |                                                                                             |                    |                                                                                     |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |

Proposal 1947 - Observation 5 - JWST Survey of the Prototypical Core-collapse Supernova Remnant Cassiopeia A

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|----------------------|-----------------------------------------------------|
| Special Requirements | Sequence Observations 4, 5, 6, 7, Non-interruptible |
|----------------------|-----------------------------------------------------|

# Proposal 1947 - Observation 6 - JWST Survey of the Prototypical Core-collapse Supernova Remnant Cassiopeia A

|                   |                                                                                             |                    |                                  |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
|-------------------|---------------------------------------------------------------------------------------------|--------------------|----------------------------------|--------|-----------------|----------------------|--------------------------|----------------|----------|-----------------|--------------------|---------------------|------------------------------|--|
| Observation       | Proposal 1947, Observation 6: MIRI IFU Knot 3                                               |                    |                                  |        |                 |                      |                          |                |          |                 |                    |                     | Thu Nov 10 23:00:46 GMT 2022 |  |
|                   | Diagnostic Status: Warning                                                                  |                    |                                  |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
|                   | Observing Template: MIRI Medium Resolution Spectroscopy                                     |                    |                                  |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
|                   | Background Observations:[MIRI IFU Background (Obs 7)]                                       |                    |                                  |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
| Diagnostics       | (MIRI IFU Knot 3 (Obs 6)) Warning (Form): Imager Filter overlap.                            |                    |                                  |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
|                   | (Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                    |                                  |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
| Fixed Targets     | #                                                                                           | Name               | Target Coordinates               |        |                 |                      | Targ. Coord. Corrections |                |          |                 | Miscellaneous      |                     |                              |  |
|                   | (9)                                                                                         | CASSIOPEIA-A-KNOT3 | RA: 23 23 30.3311 (350.8763796d) |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
|                   |                                                                                             |                    | Dec: +58 47 25.02 (58.79028d)    |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
|                   |                                                                                             |                    | Equinox: J2000                   |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
| Acquisition       | Comments:                                                                                   |                    |                                  |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
|                   | Category=ISM                                                                                |                    |                                  |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
|                   | Description=[Supernova remnants]                                                            |                    |                                  |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
|                   | Extended=YES                                                                                |                    |                                  |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
| Acquisition       | #                                                                                           | Target             |                                  |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
|                   | 1                                                                                           | NONE               |                                  |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
| Template          | AcqFilter                                                                                   | Primary Channel    |                                  |        |                 | Simultaneous Imaging |                          |                |          | Imager Subarray |                    |                     |                              |  |
|                   | F560W                                                                                       | ALL                |                                  |        |                 | YES                  |                          |                |          | FULL            |                    |                     |                              |  |
| Dithers           | #                                                                                           | Dither Type        |                                  |        |                 | Optimized For        |                          |                |          | Direction       |                    |                     |                              |  |
|                   | 1                                                                                           | 4-Point            |                                  |        |                 | EXTENDED SOURCE      |                          |                |          | NEGATIVE        |                    |                     |                              |  |
| Spectral Elements | #                                                                                           | Wavelength Range   | Detector                         | Filter | Readout Pattern | Groups/Int           | Integrations/Exp         | Exposures/Dith | Dither   | Total Dithers   | Total Integrations | Total Exposure Time | ETC Wkbk.Calc ID             |  |
|                   | 1                                                                                           |                    | IMAGER                           | F1500W | FASTR1          | 8                    | 5                        | 1              | Dither 1 | 4               | 20                 | 488.407             |                              |  |
|                   | 1                                                                                           | LONG(C)            | MRSLONG                          |        | FASTR1          | 50                   | 1                        | 1              | Dither 1 | 4               | 4                  | 555.008             |                              |  |
|                   | 1                                                                                           | LONG(C)            | MRSSHORT                         |        | FASTR1          | 50                   | 1                        | 1              | Dither 1 | 4               | 4                  | 555.008             |                              |  |
|                   | 2                                                                                           |                    | IMAGER                           | F770W  | FASTR1          | 8                    | 5                        | 1              | Dither 1 | 4               | 20                 | 488.407             |                              |  |
|                   | 2                                                                                           | MEDIUM(B)          | MRSLONG                          |        | FASTR1          | 50                   | 1                        | 1              | Dither 1 | 4               | 4                  | 555.008             |                              |  |
|                   | 2                                                                                           | MEDIUM(B)          | MRSSHORT                         |        | FASTR1          | 50                   | 1                        | 1              | Dither 1 | 4               | 4                  | 555.008             |                              |  |
|                   | 3                                                                                           |                    | IMAGER                           | F560W  | FASTR1          | 8                    | 5                        | 1              | Dither 1 | 4               | 20                 | 488.407             |                              |  |
|                   | 3                                                                                           | SHORT(A)           | MRSLONG                          |        | FASTR1          | 50                   | 1                        | 1              | Dither 1 | 4               | 4                  | 555.008             |                              |  |
|                   | 3                                                                                           | SHORT(A)           | MRSSHORT                         |        | FASTR1          | 50                   | 1                        | 1              | Dither 1 | 4               | 4                  | 555.008             |                              |  |
|                   |                                                                                             |                    |                                  |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
|                   |                                                                                             |                    |                                  |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |
|                   |                                                                                             |                    |                                  |        |                 |                      |                          |                |          |                 |                    |                     |                              |  |

Proposal 1947 - Observation 6 - JWST Survey of the Prototypical Core-collapse Supernova Remnant Cassiopeia A

|                      |                                                     |
|----------------------|-----------------------------------------------------|
| Special Requirements | Sequence Observations 4, 5, 6, 7, Non-interruptible |
|----------------------|-----------------------------------------------------|



# Proposal 1947 - Observation 7 - JWST Survey of the Prototypical Core-collapse Supernova Remnant Cassiopeia A

|                   |                                                                                                         |                   |                                                                                    |        |                 |            |                      |                          |          |                 |                    |                     |                              |
|-------------------|---------------------------------------------------------------------------------------------------------|-------------------|------------------------------------------------------------------------------------|--------|-----------------|------------|----------------------|--------------------------|----------|-----------------|--------------------|---------------------|------------------------------|
| Observation       | Proposal 1947, Observation 7: MIRI IFU Background                                                       |                   |                                                                                    |        |                 |            |                      |                          |          |                 |                    |                     | Thu Nov 10 23:00:46 GMT 2022 |
|                   | Diagnostic Status: Warning                                                                              |                   |                                                                                    |        |                 |            |                      |                          |          |                 |                    |                     |                              |
|                   | Observing Template: MIRI Medium Resolution Spectroscopy                                                 |                   |                                                                                    |        |                 |            |                      |                          |          |                 |                    |                     |                              |
|                   | Background Observation For: [MIRI IFU Knot 1 (Obs 4), MIRI IFU Knot 2 (Obs 5), MIRI IFU Knot 3 (Obs 6)] |                   |                                                                                    |        |                 |            |                      |                          |          |                 |                    |                     |                              |
| Diagnostics       | (MIRI IFU Background (Obs 7)) Warning (Form): Imager Filter overlap.                                    |                   |                                                                                    |        |                 |            |                      |                          |          |                 |                    |                     |                              |
|                   | (Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.             |                   |                                                                                    |        |                 |            |                      |                          |          |                 |                    |                     |                              |
| Fixed Targets     | #                                                                                                       | Name              | Target Coordinates                                                                 |        |                 |            |                      | Targ. Coord. Corrections |          |                 | Miscellaneous      |                     |                              |
|                   | (15)                                                                                                    | BACKGROUND-RETURN | RA: 23 24 19.8259 (351.0826079d)<br>Dec: +58 43 7.26 (58.71868d)<br>Equinox: J2000 |        |                 |            |                      |                          |          |                 |                    |                     |                              |
|                   | Comments:<br>Category=Calibration<br>Description=[Telescope/sky background]<br>Extended=YES             |                   |                                                                                    |        |                 |            |                      |                          |          |                 |                    |                     |                              |
|                   |                                                                                                         |                   |                                                                                    |        |                 |            |                      |                          |          |                 |                    |                     |                              |
| Acquisition       | #                                                                                                       | Target            |                                                                                    |        |                 |            |                      |                          |          |                 |                    |                     |                              |
|                   | 1                                                                                                       | NONE              |                                                                                    |        |                 |            |                      |                          |          |                 |                    |                     |                              |
| Template          | AcqFilter                                                                                               | Primary Channel   |                                                                                    |        |                 |            | Simultaneous Imaging |                          |          | Imager Subarray |                    |                     |                              |
|                   | F560W                                                                                                   | ALL               |                                                                                    |        |                 |            | YES                  |                          |          | FULL            |                    |                     |                              |
| Dithers           | #                                                                                                       | Dither Type       |                                                                                    |        |                 |            | Optimized For        |                          |          | Direction       |                    |                     |                              |
|                   | 1                                                                                                       | 4-Point           |                                                                                    |        |                 |            | POINT SOURCE         |                          |          | NEGATIVE        |                    |                     |                              |
| Spectral Elements | #                                                                                                       | Wavelength Range  | Detector                                                                           | Filter | Readout Pattern | Groups/Int | Integrations/Exp     | Exposures/Dith           | Dither   | Total Dithers   | Total Integrations | Total Exposure Time | ETC Wkbk.Calc ID             |
|                   | 1                                                                                                       |                   | IMAGER                                                                             | F1500W | FASTR1          | 8          | 5                    | 1                        | Dither 1 | 4               | 20                 | 488.407             |                              |
|                   | 1                                                                                                       | LONG(C)           | MRSLONG                                                                            |        | FASTR1          | 50         | 1                    | 1                        | Dither 1 | 4               | 4                  | 555.008             |                              |
|                   | 1                                                                                                       | LONG(C)           | MRSSHORT                                                                           |        | FASTR1          | 50         | 1                    | 1                        | Dither 1 | 4               | 4                  | 555.008             |                              |
|                   | 2                                                                                                       |                   | IMAGER                                                                             | F770W  | FASTR1          | 8          | 5                    | 1                        | Dither 1 | 4               | 20                 | 488.407             |                              |
|                   | 2                                                                                                       | MEDIUM(B)         | MRSLONG                                                                            |        | FASTR1          | 50         | 1                    | 1                        | Dither 1 | 4               | 4                  | 555.008             |                              |
|                   | 2                                                                                                       | MEDIUM(B)         | MRSSHORT                                                                           |        | FASTR1          | 50         | 1                    | 1                        | Dither 1 | 4               | 4                  | 555.008             |                              |
|                   | 3                                                                                                       |                   | IMAGER                                                                             | F560W  | FASTR1          | 8          | 5                    | 1                        | Dither 1 | 4               | 20                 | 488.407             |                              |
|                   | 3                                                                                                       | SHORT(A)          | MRSLONG                                                                            |        | FASTR1          | 50         | 1                    | 1                        | Dither 1 | 4               | 4                  | 555.008             |                              |
|                   | 3                                                                                                       | SHORT(A)          | MRSSHORT                                                                           |        | FASTR1          | 50         | 1                    | 1                        | Dither 1 | 4               | 4                  | 555.008             |                              |
|                   |                                                                                                         |                   |                                                                                    |        |                 |            |                      |                          |          |                 |                    |                     |                              |
|                   |                                                                                                         |                   |                                                                                    |        |                 |            |                      |                          |          |                 |                    |                     |                              |

Proposal 1947 - Observation 7 - JWST Survey of the Prototypical Core-collapse Supernova Remnant Cassiopeia A

|                      |                                                     |
|----------------------|-----------------------------------------------------|
| Special Requirements | Sequence Observations 4, 5, 6, 7, Non-interruptible |
|----------------------|-----------------------------------------------------|

# Proposal 1947 - Observation 8 - JWST Survey of the Prototypical Core-collapse Supernova Remnant Cassiopeia A

|                      |                                                                                                                                          |                    |                                                                                     |            |                  |                          |        |                  |               |                    |                     |                  |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------|------------|------------------|--------------------------|--------|------------------|---------------|--------------------|---------------------|------------------|
| Observation          | <b>Proposal 1947, Observation 8: NIRSpec Knot 1</b> <span style="float: right;">Thu Nov 10 23:00:46 GMT 2022</span>                      |                    |                                                                                     |            |                  |                          |        |                  |               |                    |                     |                  |
|                      | <b>Diagnostic Status: Warning</b><br>Observing Template: NIRSpec IFU Spectroscopy<br>Background Observations:[NIRSpec Knot Bkg (Obs 11)] |                    |                                                                                     |            |                  |                          |        |                  |               |                    |                     |                  |
| Diagnostics          | (Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                              |                    |                                                                                     |            |                  |                          |        |                  |               |                    |                     |                  |
| Fixed Targets        | #                                                                                                                                        | Name               | Target Coordinates                                                                  |            |                  | Targ. Coord. Corrections |        |                  | Miscellaneous |                    |                     |                  |
|                      | (2)                                                                                                                                      | CASSIOPEIA-A-KNOT1 | RA: 23 23 29.5212 (350.8730050d)<br>Dec: +58 50 31.34 (58.84204d)<br>Equinox: J2000 |            |                  |                          |        |                  |               |                    |                     |                  |
| Template             | <i>Comments:</i><br><i>Category=ISM</i><br><i>Description=[Supernova remnants]</i><br><i>Extended=YES</i>                                |                    |                                                                                     |            |                  |                          |        |                  |               |                    |                     |                  |
|                      | <b>TA Method</b><br>NONE                                                                                                                 |                    |                                                                                     |            |                  |                          |        |                  |               |                    |                     |                  |
| Dithers              | #                                                                                                                                        | Dither Type        |                                                                                     | Size       | Starting Point   |                          |        | Number of Points |               | Points             |                     |                  |
|                      | 1                                                                                                                                        | 4-POINT-DITHER     |                                                                                     |            |                  |                          |        |                  |               |                    |                     |                  |
| Spectral Elements    | #                                                                                                                                        | Grating/Filter     | Readout Pattern                                                                     | Groups/Int | Integrations/Exp | Leakcal                  | Dither | Autocal          | Total Dithers | Total Integrations | Total Exposure Time | ETC Wkbk.Calc ID |
|                      | 1                                                                                                                                        | G395M/F290LP       | NRSIRS2RAPID                                                                        | 10         | 1                | false                    | true   | NONE             | 4             | 4                  | 641.911             |                  |
|                      | 2                                                                                                                                        | G395M/F290LP       | NRSIRS2RAPID                                                                        | 10         | 1                | true                     | true   | NONE             | 4             | 4                  | 641.911             |                  |
| Special Requirements | Between Dates 03-OCT-2022:00:00:00 and 01-FEB-2023:00:00:00                                                                              |                    |                                                                                     |            |                  |                          |        |                  |               |                    |                     |                  |
|                      | Sequence Observations 8, 9, 10, 11, Non-interruptible                                                                                    |                    |                                                                                     |            |                  |                          |        |                  |               |                    |                     |                  |

# Proposal 1947 - Observation 9 - JWST Survey of the Prototypical Core-collapse Supernova Remnant Cassiopeia A

|                      |                                                                                                                                          |                    |                                                                                     |            |                  |         |                          |                  |               |                    |                     |                  |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------|------------|------------------|---------|--------------------------|------------------|---------------|--------------------|---------------------|------------------|
| Observation          | <b>Proposal 1947, Observation 9: NIRSpec Knot 2</b> <span style="float: right;">Thu Nov 10 23:00:46 GMT 2022</span>                      |                    |                                                                                     |            |                  |         |                          |                  |               |                    |                     |                  |
|                      | <b>Diagnostic Status: Warning</b><br>Observing Template: NIRSpec IFU Spectroscopy<br>Background Observations:[NIRSpec Knot Bkg (Obs 11)] |                    |                                                                                     |            |                  |         |                          |                  |               |                    |                     |                  |
| Diagnostics          | (Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                              |                    |                                                                                     |            |                  |         |                          |                  |               |                    |                     |                  |
| Fixed Targets        | #                                                                                                                                        | Name               | Target Coordinates                                                                  |            |                  |         | Targ. Coord. Corrections |                  |               | Miscellaneous      |                     |                  |
|                      | (8)                                                                                                                                      | CASSIOPEIA-A-KNOT2 | RA: 23 23 25.9879 (350.8582829d)<br>Dec: +58 48 43.46 (58.81207d)<br>Equinox: J2000 |            |                  |         |                          |                  |               |                    |                     |                  |
| Template             | <i>Comments:</i><br><i>Category=ISM</i><br><i>Description=[Supernova remnants]</i><br><i>Extended=YES</i>                                |                    |                                                                                     |            |                  |         |                          |                  |               |                    |                     |                  |
|                      | <b>TA Method</b><br>NONE                                                                                                                 |                    |                                                                                     |            |                  |         |                          |                  |               |                    |                     |                  |
| Dithers              | #                                                                                                                                        | Dither Type        |                                                                                     | Size       | Starting Point   |         |                          | Number of Points |               | Points             |                     |                  |
|                      | 1                                                                                                                                        | 4-POINT-DITHER     |                                                                                     |            |                  |         |                          |                  |               |                    |                     |                  |
| Spectral Elements    | #                                                                                                                                        | Grating/Filter     | Readout Pattern                                                                     | Groups/Int | Integrations/Exp | Leakcal | Dither                   | Autocal          | Total Dithers | Total Integrations | Total Exposure Time | ETC Wkbk.Calc ID |
|                      | 1                                                                                                                                        | G395M/F290LP       | NRSIRS2RAPID                                                                        | 15         | 1                | false   | true                     | NONE             | 4             | 4                  | 933.689             |                  |
|                      | 2                                                                                                                                        | G395M/F290LP       | NRSIRS2RAPID                                                                        | 15         | 1                | true    | true                     | NONE             | 4             | 4                  | 933.689             |                  |
| Special Requirements | Sequence Observations 8, 9, 10, 11, Non-interruptible                                                                                    |                    |                                                                                     |            |                  |         |                          |                  |               |                    |                     |                  |

# Proposal 1947 - Observation 10 - JWST Survey of the Prototypical Core-collapse Supernova Remnant Cassiopeia A

|                      |                                                                                                                                          |                    |                                                                                     |            |                  |                |                          |                  |               |                    |                     |                  |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------|------------|------------------|----------------|--------------------------|------------------|---------------|--------------------|---------------------|------------------|
| Observation          | <b>Proposal 1947, Observation 10: NIRSpec Knot 3</b> <span style="float: right;">Thu Nov 10 23:00:46 GMT 2022</span>                     |                    |                                                                                     |            |                  |                |                          |                  |               |                    |                     |                  |
|                      | <b>Diagnostic Status: Warning</b><br>Observing Template: NIRSpec IFU Spectroscopy<br>Background Observations:[NIRSpec Knot Bkg (Obs 11)] |                    |                                                                                     |            |                  |                |                          |                  |               |                    |                     |                  |
| Diagnostics          | (Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                             |                    |                                                                                     |            |                  |                |                          |                  |               |                    |                     |                  |
| Fixed Targets        | #                                                                                                                                        | Name               | Target Coordinates                                                                  |            |                  |                | Targ. Coord. Corrections |                  |               | Miscellaneous      |                     |                  |
|                      | (9)                                                                                                                                      | CASSIOPEIA-A-KNOT3 | RA: 23 23 30.3311 (350.8763796d)<br>Dec: +58 47 25.02 (58.79028d)<br>Equinox: J2000 |            |                  |                |                          |                  |               |                    |                     |                  |
| Template             | <i>Comments:</i><br><i>Category=ISM</i><br><i>Description=[Supernova remnants]</i><br><i>Extended=YES</i>                                |                    |                                                                                     |            |                  |                |                          |                  |               |                    |                     |                  |
|                      | <b>TA Method</b><br>NONE                                                                                                                 |                    |                                                                                     |            |                  |                |                          |                  |               |                    |                     |                  |
| Dithers              | #                                                                                                                                        | Dither Type        |                                                                                     | Size       |                  | Starting Point |                          | Number of Points |               | Points             |                     |                  |
|                      | 1                                                                                                                                        | 4-POINT-DITHER     |                                                                                     |            |                  |                |                          |                  |               |                    |                     |                  |
| Spectral Elements    | #                                                                                                                                        | Grating/Filter     | Readout Pattern                                                                     | Groups/Int | Integrations/Exp | Leakcal        | Dither                   | Autocal          | Total Dithers | Total Integrations | Total Exposure Time | ETC Wkbk.Calc ID |
|                      | 1                                                                                                                                        | G395M/F290LP       | NRSIRS2RAPID                                                                        | 15         | 1                | false          | true                     | NONE             | 4             | 4                  | 933.689             |                  |
|                      | 2                                                                                                                                        | G395M/F290LP       | NRSIRS2RAPID                                                                        | 15         | 1                | true           | true                     | NONE             | 4             | 4                  | 933.689             |                  |
| Special Requirements | Sequence Observations 8, 9, 10, 11, Non-interruptible                                                                                    |                    |                                                                                     |            |                  |                |                          |                  |               |                    |                     |                  |

# Proposal 1947 - Observation 11 - JWST Survey of the Prototypical Core-collapse Supernova Remnant Cassiopeia A

| Observation                                                                                                             | <b>Proposal 1947, Observation 11: NIRSpec Knot Bkg</b> <span style="float: right;">Thu Nov 10 23:00:46 GMT 2022</span>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                    |                    |                          |                  |         |        |         |               |                    |                     |                  |   |                |                    |                          |                  |         |                   |                                                                                    |               |                    |                                                                                                                         |                  |   |              |              |    |   |       |      |      |   |   |         |  |   |              |              |    |   |      |      |      |   |   |         |  |
|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------|--------------------------|------------------|---------|--------|---------|---------------|--------------------|---------------------|------------------|---|----------------|--------------------|--------------------------|------------------|---------|-------------------|------------------------------------------------------------------------------------|---------------|--------------------|-------------------------------------------------------------------------------------------------------------------------|------------------|---|--------------|--------------|----|---|-------|------|------|---|---|---------|--|---|--------------|--------------|----|---|------|------|------|---|---|---------|--|
|                                                                                                                         | <b>Diagnostic Status: Warning</b><br>Observing Template: NIRSpec IFU Spectroscopy<br>Background Observation For: [NIRSpec Knot 1 (Obs 8), NIRSpec Knot 2 (Obs 9), NIRSpec Knot 3 (Obs 10)]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                    |                    |                          |                  |         |        |         |               |                    |                     |                  |   |                |                    |                          |                  |         |                   |                                                                                    |               |                    |                                                                                                                         |                  |   |              |              |    |   |       |      |      |   |   |         |  |   |              |              |    |   |      |      |      |   |   |         |  |
| Diagnostics                                                                                                             | (Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                    |                    |                          |                  |         |        |         |               |                    |                     |                  |   |                |                    |                          |                  |         |                   |                                                                                    |               |                    |                                                                                                                         |                  |   |              |              |    |   |       |      |      |   |   |         |  |   |              |              |    |   |      |      |      |   |   |         |  |
| Fixed Targets                                                                                                           | <table border="1"> <thead> <tr> <th>#</th><th>Name</th><th>Target Coordinates</th><th>Targ. Coord. Corrections</th><th>Miscellaneous</th></tr> </thead> <tbody> <tr> <td>(15)</td><td>BACKGROUND-RETURN</td><td>RA: 23 24 19.8259 (351.0826079d)<br/>Dec: +58 43 7.26 (58.71868d)<br/>Equinox: J2000</td><td></td><td></td></tr> <tr> <td colspan="5"> <i>Comments:</i><br/> <i>Category=Calibration</i><br/> <i>Description=[Telescope/sky background]</i><br/> <i>Extended=YES</i> </td></tr> </tbody> </table>                                                                                                                                                                   |                                                                                    |                    |                          |                  |         |        |         |               |                    |                     |                  | # | Name           | Target Coordinates | Targ. Coord. Corrections | Miscellaneous    | (15)    | BACKGROUND-RETURN | RA: 23 24 19.8259 (351.0826079d)<br>Dec: +58 43 7.26 (58.71868d)<br>Equinox: J2000 |               |                    | <i>Comments:</i><br><i>Category=Calibration</i><br><i>Description=[Telescope/sky background]</i><br><i>Extended=YES</i> |                  |   |              |              |    |   |       |      |      |   |   |         |  |   |              |              |    |   |      |      |      |   |   |         |  |
|                                                                                                                         | #                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Name                                                                               | Target Coordinates | Targ. Coord. Corrections | Miscellaneous    |         |        |         |               |                    |                     |                  |   |                |                    |                          |                  |         |                   |                                                                                    |               |                    |                                                                                                                         |                  |   |              |              |    |   |       |      |      |   |   |         |  |   |              |              |    |   |      |      |      |   |   |         |  |
| (15)                                                                                                                    | BACKGROUND-RETURN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | RA: 23 24 19.8259 (351.0826079d)<br>Dec: +58 43 7.26 (58.71868d)<br>Equinox: J2000 |                    |                          |                  |         |        |         |               |                    |                     |                  |   |                |                    |                          |                  |         |                   |                                                                                    |               |                    |                                                                                                                         |                  |   |              |              |    |   |       |      |      |   |   |         |  |   |              |              |    |   |      |      |      |   |   |         |  |
| <i>Comments:</i><br><i>Category=Calibration</i><br><i>Description=[Telescope/sky background]</i><br><i>Extended=YES</i> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                    |                    |                          |                  |         |        |         |               |                    |                     |                  |   |                |                    |                          |                  |         |                   |                                                                                    |               |                    |                                                                                                                         |                  |   |              |              |    |   |       |      |      |   |   |         |  |   |              |              |    |   |      |      |      |   |   |         |  |
| Template                                                                                                                | <b>TA Method</b><br>NONE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                    |                    |                          |                  |         |        |         |               |                    |                     |                  |   |                |                    |                          |                  |         |                   |                                                                                    |               |                    |                                                                                                                         |                  |   |              |              |    |   |       |      |      |   |   |         |  |   |              |              |    |   |      |      |      |   |   |         |  |
| Dithers                                                                                                                 | <table border="1"> <thead> <tr> <th>#</th><th>Dither Type</th><th>Size</th><th>Starting Point</th><th>Number of Points</th><th>Points</th></tr> </thead> <tbody> <tr> <td>1</td><td>4-POINT-NOD</td><td></td><td></td><td></td><td></td></tr> </tbody> </table>                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                    |                    |                          |                  |         |        |         |               |                    |                     |                  | # | Dither Type    | Size               | Starting Point           | Number of Points | Points  | 1                 | 4-POINT-NOD                                                                        |               |                    |                                                                                                                         |                  |   |              |              |    |   |       |      |      |   |   |         |  |   |              |              |    |   |      |      |      |   |   |         |  |
|                                                                                                                         | #                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Dither Type                                                                        | Size               | Starting Point           | Number of Points | Points  |        |         |               |                    |                     |                  |   |                |                    |                          |                  |         |                   |                                                                                    |               |                    |                                                                                                                         |                  |   |              |              |    |   |       |      |      |   |   |         |  |   |              |              |    |   |      |      |      |   |   |         |  |
| 1                                                                                                                       | 4-POINT-NOD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                    |                    |                          |                  |         |        |         |               |                    |                     |                  |   |                |                    |                          |                  |         |                   |                                                                                    |               |                    |                                                                                                                         |                  |   |              |              |    |   |       |      |      |   |   |         |  |   |              |              |    |   |      |      |      |   |   |         |  |
| Spectral Elements                                                                                                       | <table border="1"> <thead> <tr> <th>#</th><th>Grating/Filter</th><th>Readout Pattern</th><th>Groups/Int</th><th>Integrations/Exp</th><th>Leakcal</th><th>Dither</th><th>Autocal</th><th>Total Dithers</th><th>Total Integrations</th><th>Total Exposure Time</th><th>ETC Wkbk.Calc ID</th></tr> </thead> <tbody> <tr> <td>1</td><td>G395M/F290LP</td><td>NRSIRS2RAPID</td><td>15</td><td>1</td><td>false</td><td>true</td><td>NONE</td><td>4</td><td>4</td><td>933.689</td><td></td></tr> <tr> <td>2</td><td>G395M/F290LP</td><td>NRSIRS2RAPID</td><td>15</td><td>1</td><td>true</td><td>true</td><td>NONE</td><td>4</td><td>4</td><td>933.689</td><td></td></tr> </tbody> </table> |                                                                                    |                    |                          |                  |         |        |         |               |                    |                     |                  | # | Grating/Filter | Readout Pattern    | Groups/Int               | Integrations/Exp | Leakcal | Dither            | Autocal                                                                            | Total Dithers | Total Integrations | Total Exposure Time                                                                                                     | ETC Wkbk.Calc ID | 1 | G395M/F290LP | NRSIRS2RAPID | 15 | 1 | false | true | NONE | 4 | 4 | 933.689 |  | 2 | G395M/F290LP | NRSIRS2RAPID | 15 | 1 | true | true | NONE | 4 | 4 | 933.689 |  |
|                                                                                                                         | #                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Grating/Filter                                                                     | Readout Pattern    | Groups/Int               | Integrations/Exp | Leakcal | Dither | Autocal | Total Dithers | Total Integrations | Total Exposure Time | ETC Wkbk.Calc ID |   |                |                    |                          |                  |         |                   |                                                                                    |               |                    |                                                                                                                         |                  |   |              |              |    |   |       |      |      |   |   |         |  |   |              |              |    |   |      |      |      |   |   |         |  |
|                                                                                                                         | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | G395M/F290LP                                                                       | NRSIRS2RAPID       | 15                       | 1                | false   | true   | NONE    | 4             | 4                  | 933.689             |                  |   |                |                    |                          |                  |         |                   |                                                                                    |               |                    |                                                                                                                         |                  |   |              |              |    |   |       |      |      |   |   |         |  |   |              |              |    |   |      |      |      |   |   |         |  |
| 2                                                                                                                       | G395M/F290LP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | NRSIRS2RAPID                                                                       | 15                 | 1                        | true             | true    | NONE   | 4       | 4             | 933.689            |                     |                  |   |                |                    |                          |                  |         |                   |                                                                                    |               |                    |                                                                                                                         |                  |   |              |              |    |   |       |      |      |   |   |         |  |   |              |              |    |   |      |      |      |   |   |         |  |
| Special Requirements                                                                                                    | Sequence Observations 8, 9, 10, 11, Non-interruptible                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                    |                    |                          |                  |         |        |         |               |                    |                     |                  |   |                |                    |                          |                  |         |                   |                                                                                    |               |                    |                                                                                                                         |                  |   |              |              |    |   |       |      |      |   |   |         |  |   |              |              |    |   |      |      |      |   |   |         |  |

# Proposal 1947 - Observation 12 - JWST Survey of the Prototypical Core-collapse Supernova Remnant Cassiopeia A

|                   |                                                                                              |                          |                                                                                     |                 |            |                  |                          |                   |               |                                                                                                                                                                                                                                                              |                     |                              |
|-------------------|----------------------------------------------------------------------------------------------|--------------------------|-------------------------------------------------------------------------------------|-----------------|------------|------------------|--------------------------|-------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------------------------------|
| Observation       | Proposal 1947, Observation 12: NIRSpec Center                                                |                          |                                                                                     |                 |            |                  |                          |                   |               |                                                                                                                                                                                                                                                              |                     | Thu Nov 10 23:00:46 GMT 2022 |
|                   | Diagnostic Status: Warning                                                                   |                          |                                                                                     |                 |            |                  |                          |                   |               |                                                                                                                                                                                                                                                              |                     |                              |
| Diagnostics       | Observing Template: NIRSpec Fixed Slit Spectroscopy                                          |                          |                                                                                     |                 |            |                  |                          |                   |               |                                                                                                                                                                                                                                                              |                     |                              |
|                   | (Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                          |                                                                                     |                 |            |                  |                          |                   |               |                                                                                                                                                                                                                                                              |                     |                              |
| Fixed Targets     | #                                                                                            | Name                     | Target Coordinates                                                                  |                 |            |                  | Targ. Coord. Corrections |                   |               | Miscellaneous                                                                                                                                                                                                                                                |                     |                              |
|                   | (11)                                                                                         | CASSIOPEIA-A-CENTER-IFU  | RA: 23 23 27.8999 (350.8662496d)<br>Dec: +58 48 37.88 (58.81052d)<br>Equinox: J2000 |                 |            |                  |                          |                   |               | Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.This object was generated by the targetselector and retrieved from the SIMBAD database.<br>Category=ISM<br>Description=[Supernova remnants]<br>Extended=YES |                     |                              |
| Template          | TA Method                                                                                    |                          | Slit                                                                                |                 |            |                  | Subarray                 |                   |               |                                                                                                                                                                                                                                                              |                     |                              |
|                   | NONE                                                                                         |                          | S1600A1                                                                             |                 |            |                  | FULL                     |                   |               |                                                                                                                                                                                                                                                              |                     |                              |
| Dithers           | #                                                                                            | Primary Dither Positions |                                                                                     |                 |            |                  |                          | Sub-Pixel Pattern |               |                                                                                                                                                                                                                                                              |                     |                              |
|                   | 1                                                                                            | 2                        |                                                                                     |                 |            |                  |                          | SPATIAL           |               |                                                                                                                                                                                                                                                              |                     |                              |
| Spectral Elements | #                                                                                            | Grating/Filter           | Slit                                                                                | Readout Pattern | Groups/Int | Integrations/Exp | #                        | Autocal           | Total Dithers | Total Integrations                                                                                                                                                                                                                                           | Total Exposure Time | ETC Wkbk.Calc ID             |
|                   | 1                                                                                            | G395M/F290LP             | S1600A1                                                                             | NRSIRS2RAPID    | 24         | 1                | 1                        | NONE              | 4             | 4                                                                                                                                                                                                                                                            | 1458.889            |                              |

# Proposal 1947 - Observation 13 - JWST Survey of the Prototypical Core-collapse Supernova Remnant Cassiopeia A

|                                    |                                                                                               |                     |                                                                                     |                    |                  |                      |                          |               |                                                                                                                                                                                                                                                              |                    |                              |
|------------------------------------|-----------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------|--------------------|------------------|----------------------|--------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|------------------------------|
| Observation                        | Proposal 1947, Observation 13: NIRCAM imaging                                                 |                     |                                                                                     |                    |                  |                      |                          |               |                                                                                                                                                                                                                                                              |                    | Thu Nov 10 23:00:46 GMT 2022 |
|                                    | Diagnostic Status: Warning                                                                    |                     |                                                                                     |                    |                  |                      |                          |               |                                                                                                                                                                                                                                                              |                    |                              |
| Observing Template: NIRCam Imaging |                                                                                               |                     |                                                                                     |                    |                  |                      |                          |               |                                                                                                                                                                                                                                                              |                    |                              |
| Diagnostics                        | (Visit 13:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.  |                     |                                                                                     |                    |                  |                      |                          |               |                                                                                                                                                                                                                                                              |                    |                              |
|                                    | (Visit 13:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.  |                     |                                                                                     |                    |                  |                      |                          |               |                                                                                                                                                                                                                                                              |                    |                              |
|                                    | (Visit 13:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.  |                     |                                                                                     |                    |                  |                      |                          |               |                                                                                                                                                                                                                                                              |                    |                              |
|                                    | (Visit 13:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.  |                     |                                                                                     |                    |                  |                      |                          |               |                                                                                                                                                                                                                                                              |                    |                              |
|                                    | (Visit 13:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.  |                     |                                                                                     |                    |                  |                      |                          |               |                                                                                                                                                                                                                                                              |                    |                              |
|                                    | (Visit 13:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.  |                     |                                                                                     |                    |                  |                      |                          |               |                                                                                                                                                                                                                                                              |                    |                              |
|                                    | (Visit 13:7) Warning (Form): Overheads are provisional until the Visit Planner has been run.  |                     |                                                                                     |                    |                  |                      |                          |               |                                                                                                                                                                                                                                                              |                    |                              |
|                                    | (Visit 13:8) Warning (Form): Overheads are provisional until the Visit Planner has been run.  |                     |                                                                                     |                    |                  |                      |                          |               |                                                                                                                                                                                                                                                              |                    |                              |
|                                    | (Visit 13:9) Warning (Form): Overheads are provisional until the Visit Planner has been run.  |                     |                                                                                     |                    |                  |                      |                          |               |                                                                                                                                                                                                                                                              |                    |                              |
|                                    | (Visit 13:10) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                     |                                                                                     |                    |                  |                      |                          |               |                                                                                                                                                                                                                                                              |                    |                              |
|                                    | (Visit 13:11) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                     |                                                                                     |                    |                  |                      |                          |               |                                                                                                                                                                                                                                                              |                    |                              |
|                                    | (Visit 13:12) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                     |                                                                                     |                    |                  |                      |                          |               |                                                                                                                                                                                                                                                              |                    |                              |
|                                    | Fixed Targets                                                                                 | #                   | Name                                                                                | Target Coordinates |                  |                      | Targ. Coord. Corrections |               |                                                                                                                                                                                                                                                              | Miscellaneous      |                              |
| (1)                                |                                                                                               | CASSIOPEIA-A-CENTER | RA: 23 23 24.0000 (350.8500000d)<br>Dec: +58 48 54.00 (58.81500d)<br>Equinox: J2000 |                    |                  |                      |                          |               | Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.This object was generated by the targetselector and retrieved from the SIMBAD database.<br>Category=ISM<br>Description=[Supernova remnants]<br>Extended=YES |                    |                              |
| Template                           | Module                                                                                        |                     |                                                                                     |                    |                  | Subarray             |                          |               |                                                                                                                                                                                                                                                              |                    |                              |
|                                    | ALL                                                                                           |                     |                                                                                     |                    |                  | FULL                 |                          |               |                                                                                                                                                                                                                                                              |                    |                              |
| Mosaic                             | Rows                                                                                          | Columns             | Row Overlap %                                                                       |                    | Column Overlap % |                      | Row shift                |               | Column shift                                                                                                                                                                                                                                                 |                    | Tile Order                   |
|                                    | 3                                                                                             | 1                   | 10.0                                                                                |                    | 10.0             |                      | 0.0                      |               | 0.0                                                                                                                                                                                                                                                          |                    | DEFAULT                      |
| Dithers                            | #                                                                                             | Primary Dither Type |                                                                                     | Primary Dithers    |                  | Subpixel Dither Type |                          | Dither Size   |                                                                                                                                                                                                                                                              | Subpixel Positions |                              |
|                                    | 1                                                                                             | FULL                |                                                                                     | 3TIGHT             |                  | STANDARD             |                          |               |                                                                                                                                                                                                                                                              | 4                  |                              |
| Spectral Elements                  | #                                                                                             | Short Filter        | Long Filter                                                                         | Readout Pattern    | Groups/Int       | Integrations/Exp     | Total Integrations       | Total Dithers | Total Exposure Time                                                                                                                                                                                                                                          | ETC Wkbk.Calc ID   |                              |
|                                    | 1                                                                                             | F162M+F150W2        | F444W                                                                               | BRIGHT1            | 7                | 1                    | 12                       | 12            | 1674.936                                                                                                                                                                                                                                                     |                    |                              |
|                                    | 2                                                                                             | F162M+F150W2        | F356W                                                                               | BRIGHT1            | 7                | 1                    | 12                       | 12            | 1674.936                                                                                                                                                                                                                                                     |                    |                              |



Proposal 1947 - Observation 13 - JWST Survey of the Prototypical Core-collapse Supernova Remnant Cassiopeia A

|                      |                                                 |
|----------------------|-------------------------------------------------|
| Special Requirements | Group Visits within 53.0 Days<br>Visits Same PA |
|----------------------|-------------------------------------------------|

# Proposal 1947 - Observation 15 - JWST Survey of the Prototypical Core-collapse Supernova Remnant Cassiopeia A

|                      |                                                                                                                                                                                                           |                       |                                                                                                                                                                                                                                                                                                                                                         |                  |                  |                          |                |                    |                    |                     |                              |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------------|--------------------------|----------------|--------------------|--------------------|---------------------|------------------------------|
| Observation          | Proposal 1947, Observation 15: MIRI Imaging additional 1                                                                                                                                                  |                       |                                                                                                                                                                                                                                                                                                                                                         |                  |                  |                          |                |                    |                    |                     | Thu Nov 10 23:00:46 GMT 2022 |
|                      | Diagnostic Status: Warning<br>Observing Template: MIRI Imaging                                                                                                                                            |                       |                                                                                                                                                                                                                                                                                                                                                         |                  |                  |                          |                |                    |                    |                     |                              |
| Diagnostics          | (Visit 15:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                              |                       |                                                                                                                                                                                                                                                                                                                                                         |                  |                  |                          |                |                    |                    |                     |                              |
| Fixed Targets        | #                                                                                                                                                                                                         | Name                  | Target Coordinates                                                                                                                                                                                                                                                                                                                                      |                  |                  | Targ. Coord. Corrections |                |                    | Miscellaneous      |                     |                              |
|                      | (12)                                                                                                                                                                                                      | CASSIOPEIA-A-1-RETURN | RA: 23 23 20.7612 (350.8365050d)<br>Dec: +58 47 37.42 (58.79373d)<br>Equinox: J2000<br><br>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.This object was generated by the targetselector and retrieved from the SIMBAD database.<br>Category=ISM<br>Description=[Supernova remnants]<br>Extended=YES |                  |                  |                          |                |                    |                    |                     |                              |
| Template             | Subarray                                                                                                                                                                                                  |                       |                                                                                                                                                                                                                                                                                                                                                         |                  |                  |                          |                |                    |                    |                     |                              |
|                      | FULL                                                                                                                                                                                                      |                       |                                                                                                                                                                                                                                                                                                                                                         |                  |                  |                          |                |                    |                    |                     |                              |
| Dithers              | #                                                                                                                                                                                                         | Dither Type           | Starting Point                                                                                                                                                                                                                                                                                                                                          | Number of Points | Points           | Starting Set             | Number of Sets | Optimized For      | Direction          | Pattern Size        |                              |
|                      | 1                                                                                                                                                                                                         | 4-Point-Sets          | 1                                                                                                                                                                                                                                                                                                                                                       | 4                |                  | 6                        | 1              | EXTENDED<br>SOURCE | POSITIVE           | DEFAULT             |                              |
| Spectral Elements    | #                                                                                                                                                                                                         | Filter                | Readout Pattern                                                                                                                                                                                                                                                                                                                                         | Groups/Int       | Integrations/Exp | Exposures/Dith           | Dither         | Total Dithers      | Total Integrations | Total Exposure Time | ETC Wkbk.Calc ID             |
|                      | 1                                                                                                                                                                                                         | F560W                 | FASTR1                                                                                                                                                                                                                                                                                                                                                  | 8                | 1                | 1                        | Dither 1       | 4                  | 4                  | 88.801              |                              |
|                      | 2                                                                                                                                                                                                         | F770W                 | FASTR1                                                                                                                                                                                                                                                                                                                                                  | 8                | 1                | 1                        | Dither 1       | 4                  | 4                  | 88.801              |                              |
|                      | 3                                                                                                                                                                                                         | F1000W                | FASTR1                                                                                                                                                                                                                                                                                                                                                  | 8                | 1                | 1                        | Dither 1       | 4                  | 4                  | 88.801              |                              |
|                      | 4                                                                                                                                                                                                         | F1130W                | FASTR1                                                                                                                                                                                                                                                                                                                                                  | 8                | 1                | 1                        | Dither 1       | 4                  | 4                  | 88.801              |                              |
|                      | 5                                                                                                                                                                                                         | F1280W                | FASTR1                                                                                                                                                                                                                                                                                                                                                  | 8                | 1                | 1                        | Dither 1       | 4                  | 4                  | 88.801              |                              |
|                      | 6                                                                                                                                                                                                         | F1800W                | FASTR1                                                                                                                                                                                                                                                                                                                                                  | 8                | 1                | 1                        | Dither 1       | 4                  | 4                  | 88.801              |                              |
|                      | 7                                                                                                                                                                                                         | F2100W                | FASTR1                                                                                                                                                                                                                                                                                                                                                  | 8                | 1                | 1                        | Dither 1       | 4                  | 4                  | 88.801              |                              |
|                      | 8                                                                                                                                                                                                         | F2550W                | FASTR1                                                                                                                                                                                                                                                                                                                                                  | 8                | 1                | 1                        | Dither 1       | 4                  | 4                  | 88.801              |                              |
| Special Requirements | Between Dates 25-OCT-2022:00:00:00 and 01-NOV-2022:23:59:59<br>Aperture PA Range 133.83544897 to 138.12544897 Degrees (V3 129.0 to 133.29)<br><br>Sequence Observations 15, 16, 17, 18, Non-interruptible |                       |                                                                                                                                                                                                                                                                                                                                                         |                  |                  |                          |                |                    |                    |                     |                              |

# Proposal 1947 - Observation 16 - JWST Survey of the Prototypical Core-collapse Supernova Remnant Cassiopeia A

|                      |                                                                                                                                                                                                           |                       |                                                                                                                                                                                                                                                                                                                                                         |                  |                  |                          |                |                    |                    |                     |                              |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------------|--------------------------|----------------|--------------------|--------------------|---------------------|------------------------------|
| Observation          | Proposal 1947, Observation 16: MIRI Imaging additional 2                                                                                                                                                  |                       |                                                                                                                                                                                                                                                                                                                                                         |                  |                  |                          |                |                    |                    |                     | Thu Nov 10 23:00:46 GMT 2022 |
|                      | Diagnostic Status: Warning<br>Observing Template: MIRI Imaging                                                                                                                                            |                       |                                                                                                                                                                                                                                                                                                                                                         |                  |                  |                          |                |                    |                    |                     |                              |
| Diagnostics          | (Visit 16:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                              |                       |                                                                                                                                                                                                                                                                                                                                                         |                  |                  |                          |                |                    |                    |                     |                              |
| Fixed Targets        | #                                                                                                                                                                                                         | Name                  | Target Coordinates                                                                                                                                                                                                                                                                                                                                      |                  |                  | Targ. Coord. Corrections |                |                    | Miscellaneous      |                     |                              |
|                      | (13)                                                                                                                                                                                                      | CASSIOPEIA-A-2-RETURN | RA: 23 23 32.8003 (350.8866679d)<br>Dec: +58 48 32.35 (58.80899d)<br>Equinox: J2000<br><br>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.This object was generated by the targetselector and retrieved from the SIMBAD database.<br>Category=ISM<br>Description=[Supernova remnants]<br>Extended=YES |                  |                  |                          |                |                    |                    |                     |                              |
| Template             | Subarray                                                                                                                                                                                                  |                       |                                                                                                                                                                                                                                                                                                                                                         |                  |                  |                          |                |                    |                    |                     |                              |
|                      | FULL                                                                                                                                                                                                      |                       |                                                                                                                                                                                                                                                                                                                                                         |                  |                  |                          |                |                    |                    |                     |                              |
| Dithers              | #                                                                                                                                                                                                         | Dither Type           | Starting Point                                                                                                                                                                                                                                                                                                                                          | Number of Points | Points           | Starting Set             | Number of Sets | Optimized For      | Direction          | Pattern Size        |                              |
|                      | 1                                                                                                                                                                                                         | 4-Point-Sets          | 1                                                                                                                                                                                                                                                                                                                                                       | 4                |                  | 6                        | 1              | EXTENDED<br>SOURCE | POSITIVE           | DEFAULT             |                              |
| Spectral Elements    | #                                                                                                                                                                                                         | Filter                | Readout Pattern                                                                                                                                                                                                                                                                                                                                         | Groups/Int       | Integrations/Exp | Exposures/Dith           | Dither         | Total Dithers      | Total Integrations | Total Exposure Time | ETC Wkbk.Calc ID             |
|                      | 1                                                                                                                                                                                                         | F560W                 | FASTR1                                                                                                                                                                                                                                                                                                                                                  | 8                | 1                | 1                        | Dither 1       | 4                  | 4                  | 88.801              |                              |
|                      | 2                                                                                                                                                                                                         | F770W                 | FASTR1                                                                                                                                                                                                                                                                                                                                                  | 8                | 1                | 1                        | Dither 1       | 4                  | 4                  | 88.801              |                              |
|                      | 3                                                                                                                                                                                                         | F1000W                | FASTR1                                                                                                                                                                                                                                                                                                                                                  | 8                | 1                | 1                        | Dither 1       | 4                  | 4                  | 88.801              |                              |
|                      | 4                                                                                                                                                                                                         | F1130W                | FASTR1                                                                                                                                                                                                                                                                                                                                                  | 8                | 1                | 1                        | Dither 1       | 4                  | 4                  | 88.801              |                              |
|                      | 5                                                                                                                                                                                                         | F1280W                | FASTR1                                                                                                                                                                                                                                                                                                                                                  | 8                | 1                | 1                        | Dither 1       | 4                  | 4                  | 88.801              |                              |
|                      | 6                                                                                                                                                                                                         | F1800W                | FASTR1                                                                                                                                                                                                                                                                                                                                                  | 8                | 1                | 1                        | Dither 1       | 4                  | 4                  | 88.801              |                              |
|                      | 7                                                                                                                                                                                                         | F2100W                | FASTR1                                                                                                                                                                                                                                                                                                                                                  | 8                | 1                | 1                        | Dither 1       | 4                  | 4                  | 88.801              |                              |
|                      | 8                                                                                                                                                                                                         | F2550W                | FASTR1                                                                                                                                                                                                                                                                                                                                                  | 8                | 1                | 1                        | Dither 1       | 4                  | 4                  | 88.801              |                              |
| Special Requirements | Between Dates 25-OCT-2022:00:00:00 and 01-NOV-2022:23:59:59<br>Aperture PA Range 133.83544897 to 138.12544897 Degrees (V3 129.0 to 133.29)<br><br>Sequence Observations 15, 16, 17, 18, Non-interruptible |                       |                                                                                                                                                                                                                                                                                                                                                         |                  |                  |                          |                |                    |                    |                     |                              |

# Proposal 1947 - Observation 17 - JWST Survey of the Prototypical Core-collapse Supernova Remnant Cassiopeia A

|                      |                                                                                                                                                                                                           |                       |                                                                                                                                                                                                                                                                                                                                                         |                  |                  |                          |                |                 |                    |                     |                              |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------------|--------------------------|----------------|-----------------|--------------------|---------------------|------------------------------|
| Observation          | Proposal 1947, Observation 17: MIRI Imaging additional 3                                                                                                                                                  |                       |                                                                                                                                                                                                                                                                                                                                                         |                  |                  |                          |                |                 |                    |                     | Thu Nov 10 23:00:46 GMT 2022 |
|                      | Diagnostic Status: Warning<br>Observing Template: MIRI Imaging                                                                                                                                            |                       |                                                                                                                                                                                                                                                                                                                                                         |                  |                  |                          |                |                 |                    |                     |                              |
| Diagnostics          | (Visit 17:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                              |                       |                                                                                                                                                                                                                                                                                                                                                         |                  |                  |                          |                |                 |                    |                     |                              |
| Fixed Targets        | #                                                                                                                                                                                                         | Name                  | Target Coordinates                                                                                                                                                                                                                                                                                                                                      |                  |                  | Targ. Coord. Corrections |                |                 | Miscellaneous      |                     |                              |
|                      | (14)                                                                                                                                                                                                      | CASSIOPEIA-A-3-RETURN | RA: 23 23 33.7392 (350.8905800d)<br>Dec: +58 50 21.01 (58.83917d)<br>Equinox: J2000<br><br>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.This object was generated by the targetselector and retrieved from the SIMBAD database.<br>Category=ISM<br>Description=[Supernova remnants]<br>Extended=YES |                  |                  |                          |                |                 |                    |                     |                              |
| Template             | Subarray                                                                                                                                                                                                  |                       |                                                                                                                                                                                                                                                                                                                                                         |                  |                  |                          |                |                 |                    |                     |                              |
|                      | FULL                                                                                                                                                                                                      |                       |                                                                                                                                                                                                                                                                                                                                                         |                  |                  |                          |                |                 |                    |                     |                              |
| Dithers              | #                                                                                                                                                                                                         | Dither Type           | Starting Point                                                                                                                                                                                                                                                                                                                                          | Number of Points | Points           | Starting Set             | Number of Sets | Optimized For   | Direction          | Pattern Size        |                              |
|                      | 1                                                                                                                                                                                                         | 4-Point-Sets          | 1                                                                                                                                                                                                                                                                                                                                                       | 4                |                  | 6                        | 1              | EXTENDED SOURCE | POSITIVE           | DEFAULT             |                              |
| Spectral Elements    | #                                                                                                                                                                                                         | Filter                | Readout Pattern                                                                                                                                                                                                                                                                                                                                         | Groups/Int       | Integrations/Exp | Exposures/Dith           | Dither         | Total Dithers   | Total Integrations | Total Exposure Time | ETC Wkbk.Calc ID             |
|                      | 1                                                                                                                                                                                                         | F560W                 | FASTR1                                                                                                                                                                                                                                                                                                                                                  | 8                | 1                | 1                        | Dither 1       | 4               | 4                  | 88.801              |                              |
|                      | 2                                                                                                                                                                                                         | F770W                 | FASTR1                                                                                                                                                                                                                                                                                                                                                  | 8                | 1                | 1                        | Dither 1       | 4               | 4                  | 88.801              |                              |
|                      | 3                                                                                                                                                                                                         | F1000W                | FASTR1                                                                                                                                                                                                                                                                                                                                                  | 8                | 1                | 1                        | Dither 1       | 4               | 4                  | 88.801              |                              |
|                      | 4                                                                                                                                                                                                         | F1130W                | FASTR1                                                                                                                                                                                                                                                                                                                                                  | 8                | 1                | 1                        | Dither 1       | 4               | 4                  | 88.801              |                              |
|                      | 5                                                                                                                                                                                                         | F1280W                | FASTR1                                                                                                                                                                                                                                                                                                                                                  | 8                | 1                | 1                        | Dither 1       | 4               | 4                  | 88.801              |                              |
|                      | 6                                                                                                                                                                                                         | F1800W                | FASTR1                                                                                                                                                                                                                                                                                                                                                  | 8                | 1                | 1                        | Dither 1       | 4               | 4                  | 88.801              |                              |
|                      | 7                                                                                                                                                                                                         | F2100W                | FASTR1                                                                                                                                                                                                                                                                                                                                                  | 8                | 1                | 1                        | Dither 1       | 4               | 4                  | 88.801              |                              |
|                      | 8                                                                                                                                                                                                         | F2550W                | FASTR1                                                                                                                                                                                                                                                                                                                                                  | 8                | 1                | 1                        | Dither 1       | 4               | 4                  | 88.801              |                              |
| Special Requirements | Between Dates 25-OCT-2022:00:00:00 and 01-NOV-2022:23:59:59<br>Aperture PA Range 133.83544897 to 138.12544897 Degrees (V3 129.0 to 133.29)<br><br>Sequence Observations 15, 16, 17, 18, Non-interruptible |                       |                                                                                                                                                                                                                                                                                                                                                         |                  |                  |                          |                |                 |                    |                     |                              |

# Proposal 1947 - Observation 18 - JWST Survey of the Prototypical Core-collapse Supernova Remnant Cassiopeia A

|                      |                                                                                                                                                                                                           |                   |                                  |                  |                  |                          |                |                    |                    |                     |                              |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|----------------------------------|------------------|------------------|--------------------------|----------------|--------------------|--------------------|---------------------|------------------------------|
| Observation          | Proposal 1947, Observation 18: MIRI Imaging additional background                                                                                                                                         |                   |                                  |                  |                  |                          |                |                    |                    |                     | Thu Nov 10 23:00:46 GMT 2022 |
|                      | Diagnostic Status: Warning                                                                                                                                                                                |                   |                                  |                  |                  |                          |                |                    |                    |                     |                              |
|                      | Observing Template: MIRI Imaging                                                                                                                                                                          |                   |                                  |                  |                  |                          |                |                    |                    |                     |                              |
|                      | Background Observation For: []                                                                                                                                                                            |                   |                                  |                  |                  |                          |                |                    |                    |                     |                              |
| Diagnostics          | (Visit 18:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                              |                   |                                  |                  |                  |                          |                |                    |                    |                     |                              |
| Fixed Targets        | #                                                                                                                                                                                                         | Name              | Target Coordinates               |                  |                  | Targ. Coord. Corrections |                |                    | Miscellaneous      |                     |                              |
|                      | (15)                                                                                                                                                                                                      | BACKGROUND-RETURN | RA: 23 24 19.8259 (351.0826079d) |                  |                  |                          |                |                    |                    |                     |                              |
|                      |                                                                                                                                                                                                           |                   | Dec: +58 43 7.26 (58.71868d)     |                  |                  |                          |                |                    |                    |                     |                              |
|                      |                                                                                                                                                                                                           |                   | Equinox: J2000                   |                  |                  |                          |                |                    |                    |                     |                              |
| Template             | Comments:<br>Category=Calibration<br>Description=[Telescope/sky background]<br>Extended=YES                                                                                                               |                   |                                  |                  |                  |                          |                |                    |                    |                     |                              |
|                      | Subarray                                                                                                                                                                                                  |                   |                                  |                  |                  |                          |                |                    |                    |                     |                              |
|                      | FULL                                                                                                                                                                                                      |                   |                                  |                  |                  |                          |                |                    |                    |                     |                              |
| Dithers              | #                                                                                                                                                                                                         | Dither Type       | Starting Point                   | Number of Points | Points           | Starting Set             | Number of Sets | Optimized For      | Direction          | Pattern Size        |                              |
|                      | 1                                                                                                                                                                                                         | 4-Point-Sets      | 1                                | 4                |                  | 6                        | 1              | EXTENDED<br>SOURCE | POSITIVE           | DEFAULT             |                              |
| Spectral Elements    | #                                                                                                                                                                                                         | Filter            | Readout Pattern                  | Groups/Int       | Integrations/Exp | Exposures/Dith           | Dither         | Total Dithers      | Total Integrations | Total Exposure Time | ETC Wkbk.Calc ID             |
|                      | 1                                                                                                                                                                                                         | F560W             | FASTR1                           | 8                | 1                | 1                        | Dither 1       | 4                  | 4                  | 88.801              |                              |
|                      | 2                                                                                                                                                                                                         | F770W             | FASTR1                           | 8                | 1                | 1                        | Dither 1       | 4                  | 4                  | 88.801              |                              |
|                      | 3                                                                                                                                                                                                         | F1000W            | FASTR1                           | 8                | 1                | 1                        | Dither 1       | 4                  | 4                  | 88.801              |                              |
|                      | 4                                                                                                                                                                                                         | F1130W            | FASTR1                           | 8                | 1                | 1                        | Dither 1       | 4                  | 4                  | 88.801              |                              |
|                      | 5                                                                                                                                                                                                         | F1280W            | FASTR1                           | 8                | 1                | 1                        | Dither 1       | 4                  | 4                  | 88.801              |                              |
|                      | 6                                                                                                                                                                                                         | F1800W            | FASTR1                           | 8                | 1                | 1                        | Dither 1       | 4                  | 4                  | 88.801              |                              |
|                      | 7                                                                                                                                                                                                         | F2100W            | FASTR1                           | 8                | 1                | 1                        | Dither 1       | 4                  | 4                  | 88.801              |                              |
|                      | 8                                                                                                                                                                                                         | F2550W            | FASTR1                           | 8                | 1                | 1                        | Dither 1       | 4                  | 4                  | 88.801              |                              |
| Special Requirements | Between Dates 25-OCT-2022:00:00:00 and 01-NOV-2022:23:59:59<br>Aperture PA Range 133.83544897 to 138.12544897 Degrees (V3 129.0 to 133.29)<br><br>Sequence Observations 15, 16, 17, 18, Non-interruptible |                   |                                  |                  |                  |                          |                |                    |                    |                     |                              |